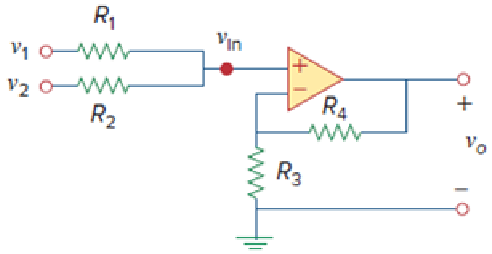
Chapter 5, Problem 34P

Problem

Given the op amp circuit shown in Fig. 5.72, express *vo* in terms of *v1* and *v2*.

Figure. 5.72:



Step-by-step solution

Step 1 of 2

Refer to Figure 5.72 in the textbook.

Apply Kirchhoff's Current Law.

$$\frac{v_1 - v_{in}}{R_1} + \frac{v_2 - v_{in}}{R_2} = 0 \quad \ldots\ldots (1)$$

Consider the voltage at the inverting terminal is v_a.

Use the voltage divider rule.

$$v_a = \frac{R_3}{R_3 + R_4} v_0 \quad \ldots\ldots (2)$$

Since,

$v_a = v_{in}$ (By the concept of virtual ground).

Step 2 of 2

Combining equations (1) and (2), we get,

$$\frac{v_1-\dfrac{R_3}{R_3+R_4}v_0}{R_1}+\frac{v_2-\dfrac{R_3}{R_3+R_4}v_0}{R_2}=0$$

$$\frac{v_1}{R_1}+\frac{v_2}{R_2}-\left(\frac{R_3}{R_3+R_4}v_0\right)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=0$$

$$\frac{v_1}{R_1}+\frac{v_2}{R_2}=\left(\frac{R_3}{R_3+R_4}v_0\right)\left(\frac{1}{R_1}+\frac{1}{R_2}\right)$$

$$v_1R_2+v_2R_1=\frac{R_3\left(R_1+R_2\right)}{R_3+R_4}v_0$$

$$v_0=\frac{R_3+R_4}{R_3\left(R_1+R_2\right)}\left(v_1R_2+v_2R_1\right)$$

$$v_0=\frac{R_3+R_4}{R_3\left(1+\dfrac{R_1}{R_2}\right)}\left(v_1+\frac{R_1}{R_2}v_2\right)$$

Therefore, the voltage, v_0 is $\boxed{\dfrac{R_3+R_4}{R_3\left(1+\dfrac{R_1}{R_2}\right)}\left(v_1+\dfrac{R_1}{R_2}v_2\right)}$.